**Mini Project**
Employee collection
Insert 50 records
Perform:

- Salary analysis
- Department headcount
- Top 5 highest paid employees
- Create index on salary
- Run aggregation pipeline

**Displaying top 3 data by using limit() function**

db["Employee"].find().limit(3)

```
> use Mydb
< switched to db Mydb
> db["Employee"].find().limit(3)
< {
    _id: ObjectId('69a6cdd51d11675756da21fa'),
    ename: 'John',
    dept: 'IT',
    salary: 800000,
    exp: 4
  }
  {
    _id: ObjectId('69a6ce8b1d11675756da21fb'),
    ename: 'Amy',
    dept: 'Support',
    salary: 250000,
    exp: 1
  }
```

```
  }
  {
    _id: ObjectId('69a6ce8b1d11675756da21fc'),
    ename: 'Rohan',
    dept: 'Sales',
    salary: 500000,
    exp: 3
  }
```

**Salary Analysis**

- Finding average salary per department
  **Query:**
  db.Employee.aggregate([{$group:{_id:"$dept",Average_salary:{$avg:"$salary"}}}])

**Output:**

```
{
  _id: 'Design',
  Average_salary: 485000
}
{
  _id: 'HR',
  Average_salary: 421818.1818181818
}
{
  _id: 'Sales',
  Average_salary: 480909.0909090909
}
{
  _id: 'IT',
  Average_salary: 589166.6666666666
}
```

- Maximum Salary per department
  **Query:**
  db.Employee.aggregate([{$group:{_id:"$dept",maximum_salary:{$max:"$salary"}}}])

**Output**

```
{
  _id: 'Design',
  maximum_salary: 750000
}
{
  _id: 'HR',
  maximum_salary: 800000
}
{
  _id: 'Support',
  maximum_salary: 320000
}
{
  _id: 'IT',
  maximum_salary: 800000
}
```

```
{
  _id: 'Sales',
  maximum_salary: 600000
}
```

- Finding Minimum Salary per department
  **Query**
  db.Employee.aggregate([{$group:{_id:"$dept",maximum_salary:{$min:"$salary"}}}])

**Output**

```
{
  _id: 'HR',
  maximum_salary: 340000
}
{
  _id: 'IT',
  maximum_salary: 390000
}
{
  _id: 'Sales',
  maximum_salary: 200000
}
{
  _id: 'Support',
  maximum_salary: 80000
}
```

```
{
  _id: 'Design',
  maximum_salary: 400000
}
```

- Total no of employees per department
  **Query**
  db.Employee.aggregate([{$group:{_id:"$dept",TotalEmployees:{$sum:1}} }])

**Output**

```
{
  _id: 'HR',
  TotalEmployees: 11
}
{
  _id: 'IT',
  TotalEmployees: 12
}
{
  _id: 'Sales',
  TotalEmployees: 11
}
{
  _id: 'Support',
  TotalEmployees: 8
}
```

```
{
  _id: 'Design',
  TotalEmployees: 8
}
```

- Displaying name and salary of top 5 high paying employees

**Query**

db.Employee.find({},{ename:1,salary:1}).sort({salary:-1}).limit(5)

**Output**

```
{
  _id: ObjectId('69a6cf561d11675756da2203'),
  ename: 'Bruno',
  salary: 800000
}
{
  _id: ObjectId('69a6cdd51d11675756da21fa'),
  ename: 'John',
  salary: 800000
}
{
  _id: ObjectId('69a6cf561d11675756da2201'),
  ename: 'Surya',
  salary: 750000
}
```

```
{
  _id: ObjectId('69a6ce8b1d11675756da21fe'),
  ename: 'Jeni',
  salary: 750000
}
{
  _id: ObjectId('69a6d16d1d11675756da2213'),
  ename: 'Faf',
  salary: 700000
}
```

- Creating index with salary column which improve our salary analysis
  **Query**
  db.Employee.createIndex({salary:1})

**Output**
Salary_1

- Average salary of people who have salary more than 500000
  **Query**
  db.Employee.aggregate([{$match:{salary:{$gt:500000}}},{$group:{_id:null,average_sal:{$avg:"$salary"}}}])

**Output**

```
{
  _id: null,
  average_sal: 633125
}
```

- Employees who having salary greater than 500000
  **Query**
  db.Employee.aggregate([{$match:{salary:{$gt:500000}}}])

**Output**

```
{
  _id: ObjectId('69a6d1ee1d11675756da221f'),
  ename: 'Arjun',
  dept: 'IT',
  salary: 520000,
  exp: 2
}
{
  _id: ObjectId('69a6d1ee1d11675756da2229'),
  ename: 'Lakshmi',
  dept: 'IT',
  salary: 530000,
  exp: 2
}
```

```
{
  _id: ObjectId('69a6cf561d11675756da2202'),
  ename: 'Rose',
  dept: 'Design',
  salary: 550000,
  exp: 3
}
{
  _id: ObjectId('69a6d0671d11675756da2210'),
  ename: 'Robert',
  dept: 'Sales',
  salary: 550000,
  exp: 2
}
```

```
{
  _id: ObjectId('69a6d16d1d11675756da221a'),
  ename: 'Susan',
  dept: 'IT',
  salary: 550000,
  exp: 3
}
{
  _id: ObjectId('69a6d1ee1d11675756da2224'),
  ename: 'Kabir',
  dept: 'IT',
  salary: 550000,
  exp: 3
}
```

- To count the no of employees per department who having experience >=5

**Query**

db.Employee.aggregate([{$match:{exp:{$gte:5}}},{$group:{_id:"$dept",Tot_Exp_Emp:{$sum:1}}}])

**Output**

```
{
  _id: 'Design',
  Tot_Exp_Emp: 1
}
{
  _id: 'HR',
  Tot_Exp_Emp: 1
}
```